



Improving uniformity and nanostructure of solution-processed thin films using ultrasonic substrate vibration post treatment (SVPT)



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ABSTRACT

The main goal of this paper is to introduce a novel mechanical method herein terms as substrate vibration post treatment (SVPT) technique, powered by ultrasonic vibration imposed on the substrate to enhance the characteristics and functionality of spun-on thin films or thin films made by similar casting techniques, such as drop and dip coating. In this technique, the as-casted wet films are placed on a substrate vibrated by an ultrasonic transducer with controlled power and duration to improve the film characteristics, such as uniformity and nanostructure. The performance of this technique is examined on spun-on PEDOT: PSS thin films used in polymer and perovskite solar cells and unprecedented results are presented. We first explore the influence of the vibration duration time on the characteristics of the films made by pristine PEDOT: PSS solution, where it is found that the optimized vibration duration for the pristine PEDOT: PSS film is about 10 s, resulting in significant increase in the film electrical conductivity and lowered thickness and roughness. In order to further test the generality and merit of the method, thin films made using PEDOT: PSS solution modified with various types of surfactants and cured by the SVPT are studied. The results show that the application of the SVPT method combined with surfactant modification leads to an impressive twelve-fold increase in the conductivity of the PEDOT: PSS thin films compared with that of the pristine non-vibrated PEDOT: PSS thin films. The sole effect of the SVPT is a four-fold increase in the conductivity of pristine PEDOT: PSS film compared with that of the non-vibrated film. This remarkable enhancement in conductivity is further explained by the AFM phase images of PEDOT: PSS films, showing that the ultrasonic energy could loosen the Coulomb forces between PEDOT and PSS chains, resulting in phase separation and localized reordering of the conducting PEDOT chains leading to an increase in the electrical conductivity of the film. Highly conductive PEDOT: PSS thin film is a viable candidate as electrodes in emerging solution-processed solar cells.

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1. Introduction

Conducting polymers have been widely used in the fields of sensors, microelectronics, flexible displays and energy conversion and storage devices, owing to their excellent electronic, optical and mechanical properties, as well as their low-cost and sustainability [1,2]. Since the synthesis of high conductivity polyacetylene by Heeger and coworkers [3], a variety of conducting polymers have been synthesized for use in emerging nanotechnologies, such as organic solar cells. However, these conducting polymers are usually insoluble in water resulting in poor process-ability [4]. One widely-used conducting co-polymer is poly(3,4-ethylenedioxy thiophene) (PEDOT): poly(4-styrenesulfonate) (PSS) or simply,

PEDOT: PSS. PEDOT is a positively-charged conducting polymer and a polythiophene derivative, which is insoluble in water. To tackle the insolubility problem, the negatively-charged PSS is blended with PEDOT, helping dispersion and stabilization of PEDOT in water and other favored solvents [5]. PEDOT: PSS is one of the most successful conducting polymers commercially produced in large scale for energy conversion devices [6]. The microstructure of PEDOT: PSS presents a loosely cross-linked and highly swollen polymer gel network in which the molecular weight of PEDOT segments is very small resulting in rather short chain lengths compared with the soluble PSS counterpart [7,8]. This unique structure makes PEDOT: PSS possess several advantages, such as having good stability against light and electricity in air, being transparent in the visible light, showing excellent solubility, process-ability, and exhibiting controllable electrical conductivity from 10^{-2} to 10^3 S/cm, to name a few [5]. However, owing to the rapid development in organic photovoltaic and photoelectronic

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devices, there is a huge demand for new generation of PEDOT: PSS with higher conductivity. To address this issue, various methods have been developed focusing on the conductivity enhancement of PEDOT: PSS [9].

The solvent treatment method has been so far proved to be the most effective approach for enhancing the conductivity of PEDOT: PSS [10,11]. A post-treated PEDOT: PSS film has been reported by fumed H_2SO_4 with a world-recording conductivity of 4380 S/cm [12]. This is because the concentrated sulfuric acid treatment results in removal of the insulating PSS chains and rearrangement of the crystallized PEDOT nanofibrils, forming a conducting network. Another remarkable organic solvent is the methanesulfonic acid that has shown to increase the conductivity of pristine PEDOT: PSS from 0.3 S/cm to 3300 S/cm, presumably due to the proton transfer from the mild organic acid to PSS^- of PEDOT: PSS [13]. The surfactant treatment method of PEDOT: PSS can also effectively enhance the conductivity and as a secondary desirable modification, it may result in reduced contact angle [14]. Jeong et al. [15] reported that a non-ionic surfactant, Triton X-100, could significantly improve the conductivity of commercial PEDOT: PSS to 100 S/cm by inducing the formation of PEDOT nanofibrils. Cao et al. [16] introduced the PEDOT: PSS/surfactant bilayer transparent electrodes with conductivity of 1019 S/cm by forming an interpenetrating conducting network.

Despite their success, the above-mentioned chemical methods have intrinsic deficiency, because excessive amount of chemicals is wasted and the solvents are not environmentally friendly; for instance, the dangerous concentrated sulfuric acid used for the film treatment may be harmful to the users and environment. Thus, the application of physical and mechanical methods for improving the conductivity of PEDOT: PSS film are appealing alternatives to the chemical methods described above. The thermal treatment, as a physical treatment method, has a great influence on the conductivity of PEDOT: PSS films through affecting the polymer nanostructure [17]. A change in the film conductivity, and therefore, electric and electronic properties of the film, alters the thin film device performance [18]. Recently, we have developed a novel spray coating technique, a mechanical treatment method, termed as the substrate vibration-assisted spray coating (SVASC). The method employs a mild and controlled ultrasonic vibration imposed on the substrate that is subjected to spray coating to improve droplet spreading, coalescence, and wetting. The SVASC was used to fabricate spray-on PEDOT: PSS thin films, where it was shown that the imposed ultrasonic vibration contributes to the film surface uniformity and four times increase in the conductivity of PEDOT: PSS films [19]. The SVASC has been used to fabricate PEDOT: PSS film on paper substrates as well, to pave the way for the fabrication of paper-based solar cells [20]. The idea of imposing ultrasonic vibration on the substrate has been tested successfully on drop-casted thin films of PEDOT: PSS used in an inverted perovskite solar cell as well, in a method called, ultrasonic substrate vibration-assisted drop casting (SVADC) [21]. While mild and controlled ultrasonic vibration power improves the film nanostructure and surface characteristics, an excessively powerful vibration, on the other hand, may result in film instability and dewetting and film rupture [22].

Inspired by the SVASC and SVADC coating methods described above, in which ultrasonic vibration is imposed on a substrate which is subjected to spray or drop coating, respectively, to achieve significant improvement in film conductivity and uniformity, here we propose and develop a novel method in which substrate vibration is used as a mechanical post treatment step applied to wet spin-coated films. The imposed substrate vibration, in principle, is applicable to other casting methods as well, such as dip coating. The method herein is termed as the ultrasonic substrate vibration post treatment (SVPT) of wet spin-coated films. The low amplitude

vibration created by a piezoelectric ultrasonic transducer attached beneath the substrate can affect the nanostructure and morphology of the wet spun-on film by realigning the molecules or polymer chains in the film. In this paper, the SVPT is applied on wet spun-on PEDOT: PSS films, where it is shown that this process results in significant increase in the film conductivity, an important functional property of the film for thin film device fabrication. We examine the effect of the substrate vibration and duration of vibration on the conductivity and morphology of pristine and chemically modified PEDOT: PSS films using three surfactants. Then we show unprecedented results affirming that the PEDOT: PSS conductivity is enhanced for both pristine and chemically-modified solution of PEDOT: PSS, indicating the great potential of the SVPT method as a universal post treatment method for the fabrication of highly conductive PEDOT: PSS and similar films.

The SVPT process followed in this work is depicted in Fig. 1. According to Fig. 1a, the wet PEDOT: PSS film is fabricated by spin coating and immediately is transferred and placed atop a piezoelectric transducer mounted inside of a metal box. The wet film is ultrasonically agitated by the high frequency of vibration of the transducer and its nanostructure is affected and finally cured and dried. Fig. 1b illustrates the commercial PEDOT: PSS solution treated with three different kinds of surfactants and their structural formulas. In the rest of this paper, details of materials preparation and experimental procedure, major results on the performance of the SVPT applied to PEDOT: PSS films are presented and discussed.

2. Materials and methods

Pristine PEDOT: PSS aqueous solution (Sigma–Aldrich, USA) was used as received with 1.3 wt.% of commercial PEDOT: PSS (0.5% PEDOT: 0.8% PSS) dispersed in water. The solution has an electrical conductivity of 1 S/cm. The anion surfactant $\text{C}_{24}\text{H}_{45}\text{NO}_6\text{S}$ (CAS No.: 27323-41-7), the cation surfactant $\text{C}_{22}\text{H}_{48}\text{N}_2\text{O}_4$ (CAS No.: 86443-82-5), and the non-ionic surfactant $\text{C}_{14}\text{H}_{22}\text{O}(\text{C}_2\text{H}_4\text{O})_n$ (CAS No.: 9002-93-1) (Chongchuan District Zhen An Chemical Additives Business Department, China) were used to modify PEDOT: PSS in selected experiments. The structural formulas of the aforementioned surfactants are shown in Fig. 1b. The substrates (bare glass, 25 mm $\times$ 25 mm) were cleaned in an ultrasonic bath by detergent, ethanol (99%, analytical grade, Sigma–Aldrich, USA), and deionized water for 15 min, respectively, then dried in a vacuum furnace and cleaned in an ultraviolet cleaner for 15 min.

The modified PEDOT: PSS (m-PEDOT: PSS) solutions were prepared by treating the pristine PEDOT: PSS by the aforementioned surfactants. In the case of the anion surfactant treatment, we added the anion surfactant into the pristine PEDOT: PSS solution at a weight ratio of 1% at room temperature and then stirred the solution using a magnetic stirrer for at least 12 h. In the rest of this paper, the PEDOT: PSS solution treated by the anion surfactant is referred to as a-PEDOT: PSS. Repeating the same process with cation and non-ionic surfactants the c-PEDOT: PSS and n-PEDOT: PSS solutions are obtained, respectively. The reference pristine PEDOT: PSS solution is simply referred to as PEDOT: PSS. The as-prepared solutions all exhibited dark blue color¹.

All four types of the solutions mentioned above were used to fabricate thin films based on the same process parameters and the following fabrication procedure. Five drops of the precursor solution were transferred onto a clean glass substrate placed on the holder of a spin coater. Using a spinning speed of 2500 rpm for 25 s, a spun-on film formed. For the samples dried under

¹ For interpretation of color in Figs. 1–9, the reader is referred to the web version of this article.

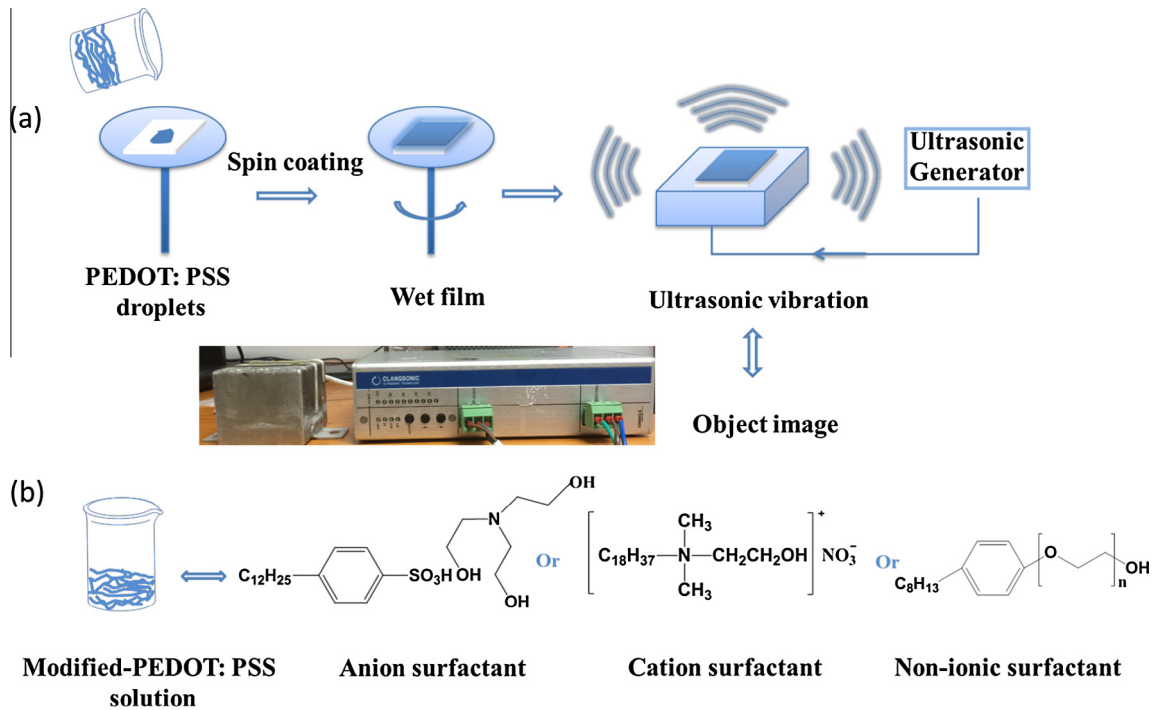


Fig. 1. (a) The film fabrication steps using spin coating followed by the SVPT. The photograph shows the ultrasonic transducer and the function generator; (b) structural formulas of three different kinds of surfactants used in this study.

imposed vibration, the as-prepared wet spun-on films were immediately placed on an ultrasonically vibrating box followed by an annealing step on a hot plate at 120 °C for 20 min. The ultrasonic transducer was installed in a metal box and used as a platform to hold and transfer ultrasonic vibrations to glass substrates to improve the coating characteristics. The ultrasonic substrate vibration was generated using a signal generator (Clangsonic, SONOCLG MS600, China) at a power of 5 W, and a fixed frequency of 40 kHz. The transducer generates vibrations in all three dimensions; the vertical displacement or amplitude of vibration was measured to be 8.7 μm, using a vibro-meter. For the non-vibrated samples, the wet spun-on films were directly annealed on the hot plate. For the pristine PEDOT: PSS films, the samples were vibrated at the following durations: 0, 10, 20, 30, and 40 s; the samples prepared are denoted as PEDOT: PSS-N, PEDOT: PSS-Y10, PEDOT: PSS-Y20, PEDOT: PSS-Y30 and PEDOT: PSS-Y40, respectively, where N stands for “no vibration applied”, and Y stands for “vibration applied”. For the modified m-PEDOT: PSS films, the films were vibrated for 10 s and are denoted as a-PEDOT: PSS-Y, c-PEDOT: PSS-Y, and n-PEDOT: PSS-Y, for anionic, cationic, and non-ionic surfactants, respectively. The non-vibrated counterparts for those films are denoted as a-PEDOT: PSS-N, c-PEDOT: PSS-N, and n-PEDOT: PSS-N.

Surface profiles, topography images, surface histogram, and film global roughness and film thicknesses of pristine PEDOT: PSS and m-PEDOT: PSS films were obtained by a nanoscale resolution confocal laser scanning microscope (CLSM, Zeiss, model LMS700, Germany). In order to measure the thickness, ethanol was used to wash off and expose the edge of the film to construct a height difference between the film and the substrate. The local roughness, morphology and phase images were obtained by an atomic force microscope (AFM, model Dimension 3100, Veeco Instruments, Inc., USA). The *I*-*V* curves of pristine PEDOT: PSS and m-PEDOT: PSS films were obtained by the 4-point probe measurement technique (Ecopia, model HSM-5000, Four Point Probes/Bridge Technology, USA). The electrical contacts were made by dropping

silver paste on the four corners of the 25 mm × 25 mm polymer films spun on glass substrates. The electrical conductivity (σ), which is the reciprocal of resistivity (ρ), was calculated through the Van der Pauw method [23], based on the following equation:

$$\sigma = \frac{1}{\rho} = \frac{\ln 2}{\pi d} \left(\frac{I}{V} \right) \quad (1)$$

where d is the film thickness, V is the voltage and I is the current.

X-ray diffraction (XRD) crystallography patterns of selected PEDOT: PSS thin films fabricated with and without applying substrate vibration were obtained using an XRD analyzer (D8-FABLINE, Bruker, USA).

3. Results and discussion

3.1. The influence of SVPT on pristine PEDOT: PSS thin films

In the SVPT method, the vibration power and duration are expected to influence the film characteristics, just similar to post annealing, in which the annealing temperature and duration affect the film characteristics. Thus, to investigate the effect of vibration duration, the spun-on pristine PEDOT: PSS films were subjected to the SVPT with various duration times at a fixed low vibration power of 5 W. Higher vibration powers may make the wet film unstable due to the generation of large amplitude surface waves [22], and therefore were not tested here. Table 1 lists the operating conditions, measured thickness and roughness values of spun-on PEDOT: PSS thin films, with different duration times. Our previous works substantiate the accuracy of the CLSM for obtaining the film thickness, particularly for the spun-on films which are smooth and uniform [24]. Thus, here the low-cost, fast, and non-intrusive CLSM method was employed to measure the film thickness.

Fig. 2 shows the surface profiles along an arbitrary line and the method of the film thickness measurement with various vibration durations using laser images taken by the CLSM. The laser images

Table 1

The thickness, global roughness and conductivity of spun-on PEDOT: PSS thin films subjected to SVPT method using different vibration duration times. For all cases, annealing temperature = 120 °C; annealing time = 20 min; spinning speed = 2500 rpm; spinning time = 20 s. Scanning surface area by CLSM: 4098 μm^2 (64 $\mu\text{m} \times 64 \mu\text{m}$).

Sample ID code	Vibration duration (s)	Film thickness (nm)	Roughness measured by CLSM (nm)	Conductivity (S/cm)
PEDOT: PSS-N	0	110	24	0.14
PEDOT: PSS-Y10	10	55	14	0.54
PEDOT: PSS-Y20	20	80	19	0.33
PEDOT: PSS-Y30	30	60	29	0.27
PEDOT: PSS-Y40	40	130	31	0.10

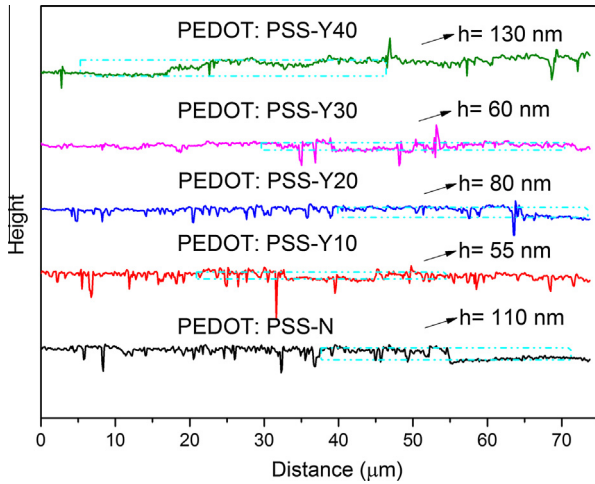


Fig. 2. Surface profiles along an arbitrary line and the method of the film thickness measurement with different vibration duration using laser images taken by the CLSM.

are shown in the Supporting Information, Fig. S1. As shown in Fig. 2 and Table 1, the non-vibrated PEDOT: PSS film has a thickness of 110 nm, while a 10 s substrate vibration results in a significant decrease in the film thickness down to 55 nm, which is the lowest film thickness among all five samples of Fig. 2. At the vibration duration of 20 and 30 s, the film thickness is 80 and 60 nm, respectively, again below the initial thickness of the non-vibrated film. However, the film thickness of PEDOT: PSS vibrated for 40 s increases to 130 nm. Based on the thickness data of Table 1, it can be argued that a short duration (~ 10 s) ultrasonic vibration results in a decrease in film thickness, while a long duration ultrasonic vibration, for instance, 40 s, leads to the formation of thicker films, perhaps due to the film break up and accumulation of some vibrated parts of the wet film, as a result of the application of excessive ultrasonic energy. A similar conclusion can be drawn by inspecting the effect of vibration duration on the film roughness data given in Table 1, i.e., substrate vibration for 10 s yields the most uniform film.

Fig. 3 shows the surface topography and histograms of five samples fabricated using pristine PEDOT: PSS, at various vibration duration times. A histogram is a graphical representation of the distribution of numerical data. It is an estimate of the probability distribution of a continuous variable. Here we used the root mean square value of the ordinates within the sample area (RSq) to define the film roughness value. To better represent the surface roughness, this type of roughness is calculated over an area instead of over a single line profile. The capital “S” in the term “RSq” is to indicate that the roughness has been calculated over an area [25]. The skewness of topography height distribution (RSsk) is the ratio of the mean cube value of the height values and the cube of RSq within a sampling area, and the kurtosis of topography height distribution (RSku) is defined as the ratio of the mean of the fourth power of the height values and the fourth power of RSq within the sampling area. RSsk and RSku are defined as follows [26]:

$$RSsk = \frac{1}{RSq^3} \left[\frac{1}{A} \right] \iint z^3(x,y) dx dy \quad (2)$$

$$RSku = \frac{1}{RSq^4} \left[\frac{1}{A} \right] \iint z^4(x,y) dx dy \quad (3)$$

where A is the measured area and $z(x,y)$ is the height of the measured points. The skewness and kurtosis values show the asymmetry of surface deviations about the reference plane and the sharpness or peakedness of the surface height distribution, respectively. For the Gaussian surface, the skewness is zero due to the symmetrical topography. If the RSsk is greater than 0, the measured surface shape has peaks or ridges. Similarly, a negative RSsk denotes a flat plane with valleys or pits. A greater RSsk corresponds to a more asymmetric surface. As to the kurtosis, the Gaussian surface has a value of RSku equal to 3. If the surface RSku is greater than 3, the surface is spiky showing a centrally distributed surface whereas RSku of smaller than 3 is indicative of a curvy surface. By a combination of RSq with RSsk and RSku, it may be possible to identify the surface shape in a physical sense [27].

The RSq, RSsk, and RSku values of each sample are shown in Fig. 3. For the PEDOT: PSS-N sample, the RSq is 36 nm while for the PEDOT: PSS-Y10 and PEDOT: PSS-Y20 films, the RSq decreases, respectively to 29 nm and 27 nm. However, the RSq values of PEDOT: PSS-Y30 and PEDOT: PSS-Y40 samples both increase to around 38 nm, showing that excessive ultrasonic energy could be detrimental to the roughness of the PEDOT: PSS film. The RSsk and RSku values can further unveil the surface shapes. For pristine PEDOT: PSS film, its surface shape shows sharp peaks and spiky surface based on the RSsk = 0.62 and RSku = 4.6, while for the PEDOT: PSS-Y10 and PEDOT: PSS-Y20, their RSsk values are respectively 0.26 and 0.66, indicating that PEDOT: PSS-Y10 has a more symmetric surface compared with the pristine PEDOT: PSS film and PEDOT: PSS-Y20. For the kurtosis index, since all samples show RSku values larger than 3, they all present spiky surfaces which are also shown in the surface topography images in the left column. For the excessively vibrated samples, i.e., PEDOT: PSS-Y30 and PEDOT: PSS-Y40, the skewness is negative denoting that the films have valleys induced by the excessive vibration.

The ratio of I over V (I/V , inverse of resistance, c.f. Fig. S2) of spun-on PEDOT: PSS thin films treated by the SVPT increases as the vibration duration increases from 0 to 30 s, reaching the highest value when the vibration lasts for 10 s. However, the I/V value decreases to a value below that of the non-vibrated sample when the vibration duration is 40 s. Since the I/V value varies with the film thickness, the film or sheet conductivity, which is an intrinsic property of the film, is calculated using Eq. (1) and shown in Fig. 4 and Table 1. It is observed that the conductivities vary with the vibration duration time. For the non-vibrated sample, the four-point sheet conductivity is as low as 0.14 S/cm, whereas the conductivity of PEDOT: PSS-Y10 film increases to 0.54 S/cm, showing four to five times increase compared with the non-vibrated sample. However, as the vibration duration further increases, the conductivity gradually decreases and becomes 0.1 S/cm at the vibration duration of 40 s, which is lower than that of the non-vibrated sample, thus proving that extending the vibration time

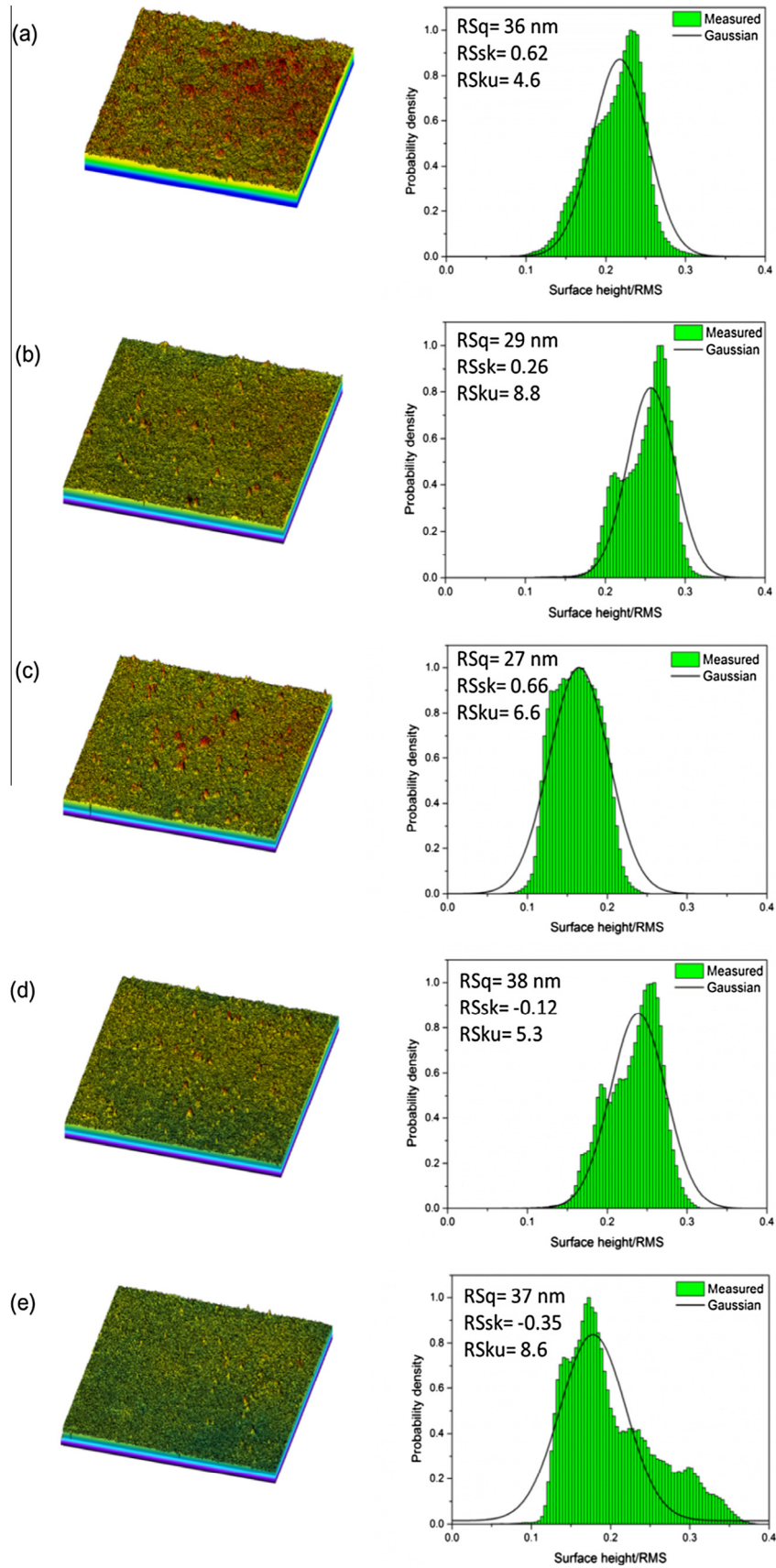


Fig. 3. The surface topography images and histograms of (a) PEDOT: PSS-N, (b) PEDOT: PSS-Y10, (c) PEDOT: PSS-Y20, (d) PEDOT: PSS-Y30, and (e) PEDOT: PSS-Y40.

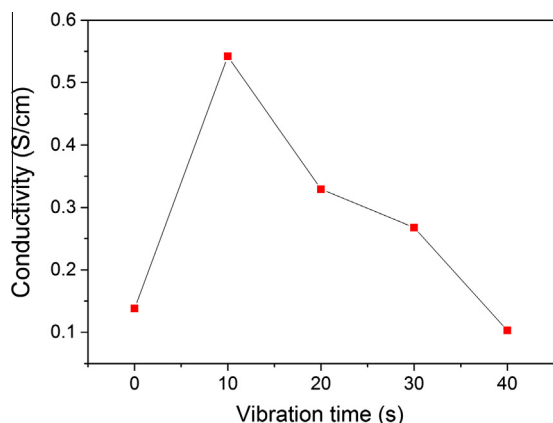


Fig. 4. The four-point electrical conductivity of pristine PEDOT: PSS thin films with different vibration duration. The 10 s vibration duration time results in the highest conductivity.

beyond a critical value results in a decline in the electrical conductivity of PEDOT: PSS films. This is because the continuous long-duration ultrasonic energy imparted to the film may cause dewetting in PEDOT: PSS film, while short-duration ultrasonic vibration can significantly enhance the conductivity by rearranging the inner structure of the wet PEDOT: PSS film.

In summary, in the forgoing section we investigated the influence of the substrate vibration duration on the thickness, roughness and conductivity of pristine PEDOT: PSS films. It is manifested that in the range of the parameters studied here and for an ultrasonic vibration power of 5 W, the optimized vibration time duration for the treatment of PEDOT: PSS films using the SVPT method is 10 s, simultaneously rendering the lowest thickness, roughness, and the highest conductivity among all samples. Increasing the vibration time beyond 10 s can lead to unfavored results, such as increased roughness and decreased conductivity, which may limit the application of the film in a device such as thin film solar cells.

3.2. The influence of SVPT on surfactant-modified PEDOT: PSS films

In order to further investigate the merit, generality, and mechanism of operation of the SVPT method, we modified the pristine PEDOT: PSS by different kinds of surfactants, i.e., anion, cation, and non-ionic surfactants, as depicted in Fig. 1b. The film thickness, local roughness and conductivity of the modified films fabricated with and without applying the SVPT were then compared to disclose the mechanism of operation of SVPT by analyzing the AFM phase images of the films. In this case, the imposed vibration was applied for 10 s, the optimum duration time obtained for pristine PEDOT: PSS films in the previous section. The results are listed in Table 2.

According to Fig. 5 and Table 2, the reference pristine PEDOT: PSS-N film has the highest thickness of 110 nm. After surfactant treatment, the film thickness decreases, where the least thickness is achieved for the film treated by the cation surfactant $C_{22}H_{48}N_2O_4$ with the thickness of 70 nm. With the application of the SVPT method on the surfactant-modified films, the thickness further decreases compared with their non-vibrated counterparts, showing the least roughness of 40 nm for the film modified by the anion surfactant. The laser surface images of m-PEDOT: PSS films used to measure the film thickness may be found in Fig. S3. Fig. 5b shows the trend of variation of thickness of m-PEDOT: PSS films with and without employing the SVPT, substantiating that post-vibration can effectively reduce the thickness of the surfactant-modified films.

Table 2

The thickness, local roughness, and conductivity of spun-on m-PEDOT: PSS thin films treated by three different kinds of surfactants at the vibration duration of 10 s. In all cases: the annealing temperature = 120 °C; annealing time = 20 min; spinning speed = 2500 rpm; spinning time = 20 s; scanning surface area by AFM = 0.25 μm^2 (0.5 $\mu\text{m} \times 0.5 \mu\text{m}$).

Sample ID code	Film thickness (nm)	Roughness measured by AFM (nm)	Conductivity (S/cm)
PEDOT: PSS-N	110	1.3	0.14
PEDOT: PSS-Y	55	1.1	0.54
a-PEDOT: PSS-N	100	1.7	0.68
a-PEDOT: PSS-Y	40	1.4	1.70
c-PEDOT: PSS-N	70	1.0	0.34
c-PEDOT: PSS-Y	60	1.0	0.67
n-PEDOT: PSS-N	100	1.8	0.10
n-PEDOT: PSS-Y	60	1.5	0.21

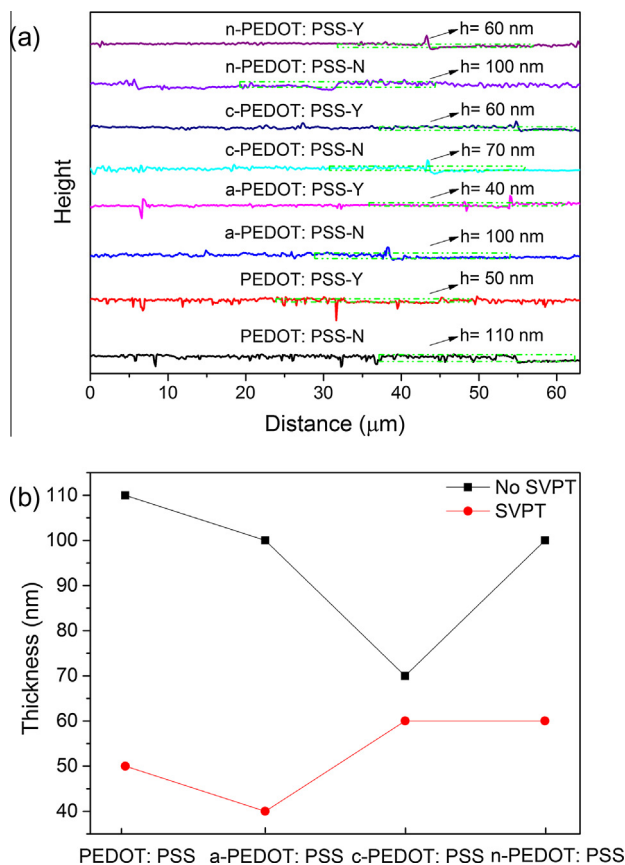


Fig. 5. (a) Surface profiles along an arbitrary line and the method of film thickness measurement for m-PEDOT: PSS films made using various surfactants with and without vibration; (b) the variation trend of the film thickness values made with and without the SVPT.

Fig. 6a shows the AFM images of the surface topography and AFM-measured local roughness of the m-PEDOT: PSS films. In Fig. 6a, the surfaces of the m-PEDOT: PSS films with employing the SVPT method appear more uniform compared with those without using the SVPT. To investigate the local surface roughness (measured over a small area), the roughness values were measured by the AFM. For those samples without employing the SVPT, the local roughness values are slightly higher. For the pristine PEDOT: PSS-N film, the roughness is around 1.3 nm. After modification with the anion and non-ionic surfactants, the roughness increases, but it decreases when the cation surfactant is used. With the application of the SVPT, the roughness values of all films decrease with

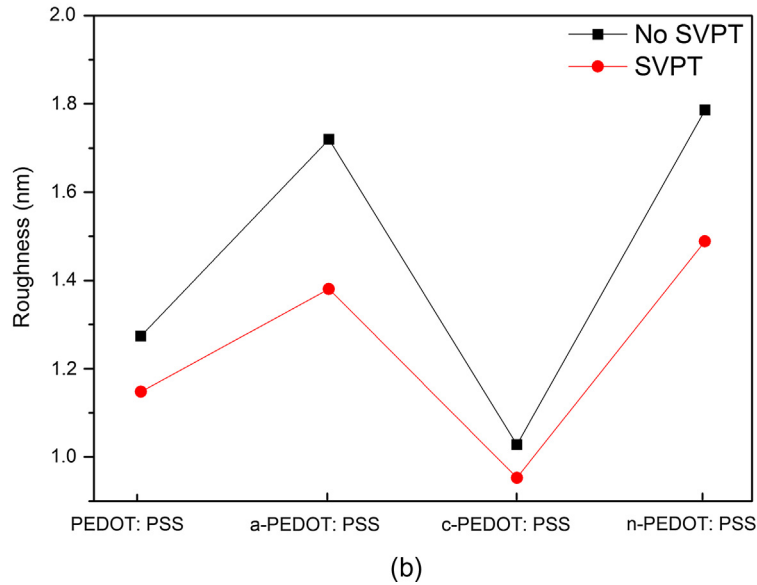
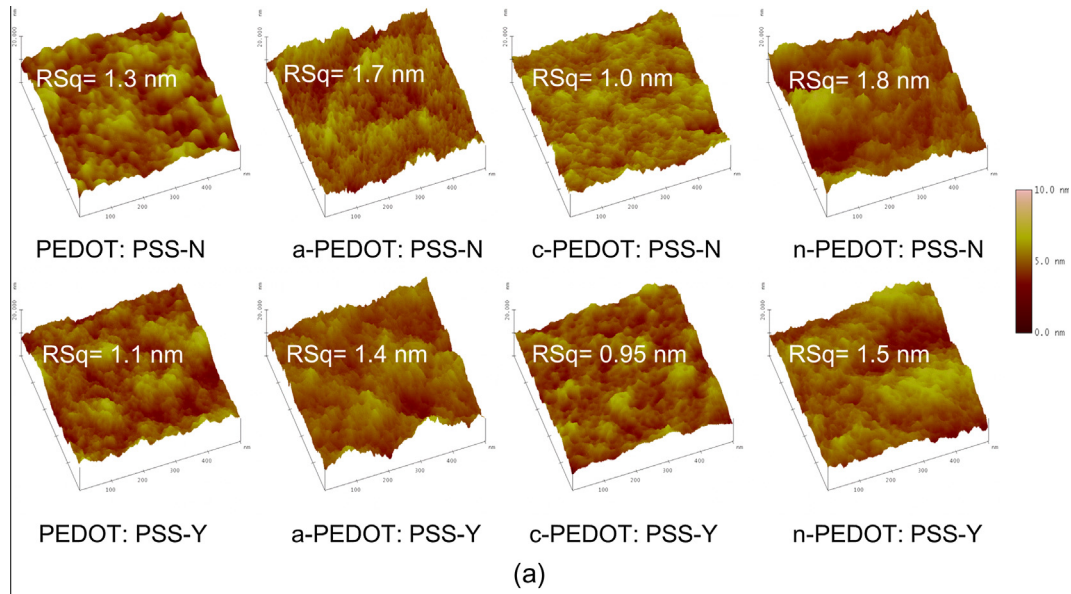


Fig. 6. (a) The AFM height images of m-PEDOT: PSS films without SVPT (upper row) and with SVPT (lower row); (b) the variation of the film roughness for various m-PEDOT: PSS films made with and without SVPT. In part (a), the AFM images are $0.5 \mu\text{m} \times 0.5 \mu\text{m}$; the RSq roughness values are also shown on the AFM images.

the least value of 0.95 nm. Fig. 6b clearly illustrates the trend of variation of roughness under the influence of using various surfactants and imposed vibration. The order of magnitude of our measured roughness values are well-matched with those reported in the literature [16,28]. It is noted that for the pristine PEDOT: PSS films, the roughness was measured over a large area of $4098 \mu\text{m}^2$, using the CLSM laser images (Table 1), while for the surfactant-modified films, the roughness was measured by AFM over a small area of $0.25 \mu\text{m}^2$ (Table 2). It is evident that the local AFM roughness is much smaller than the global CLSM roughness.

Fig. 7 illustrates the conductivity of the m-PEDOT: PSS films based on the I/V values (reported in Fig. S4) and thickness of the spun-on m-PEDOT: PSS films. It is observed that the sheet conductivity of the samples treated by SVPT increases compared with that of the non-vibrated samples. For the pristine PEDOT: PSS-N sample (without vibration), the conductivity is 0.14 S/cm. After modification with anion and cation surfactants, the conductivity increases to 0.68 and 0.34 S/cm, respectively, while the non-ionic

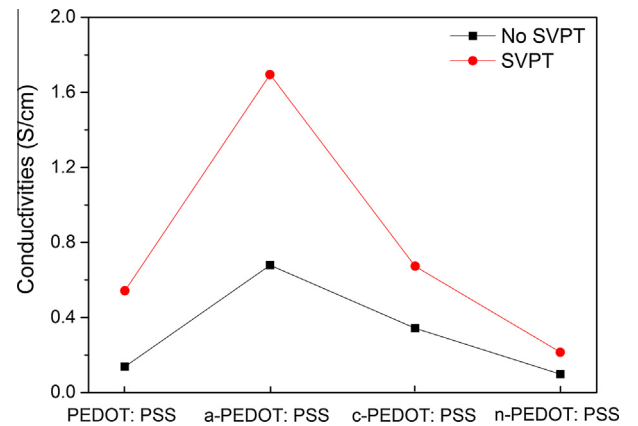


Fig. 7. The four-point electrical conductivity of the m-PEDOT: PSS films made with and without applying the SVPT.

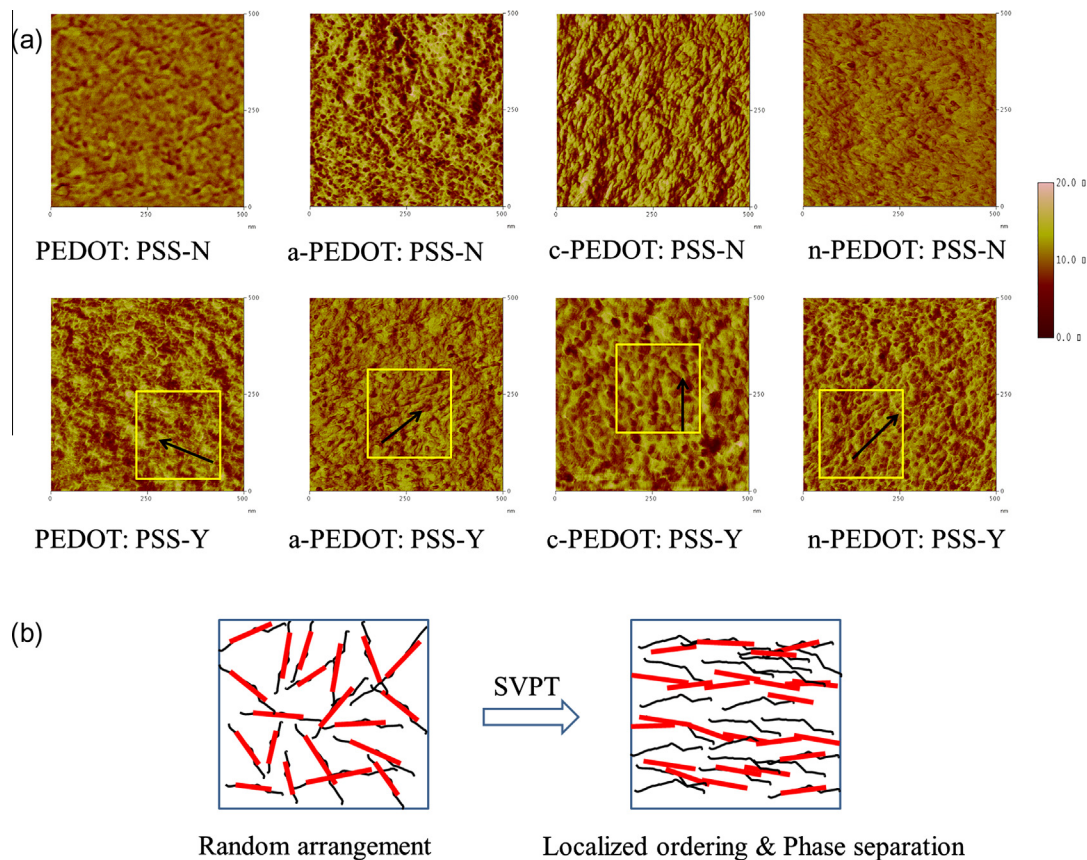


Fig. 8. (a) AFM phase images of m-PEDOT: PSS films without SVPT (upper row) and with SVPT (lower row), and (b) schematic of the proposed mechanism of SVPT (the short thick lines represent the PEDOT chains and the long thin lines denote the PSS chains). In part (a), the AFM images are $0.5 \mu\text{m} \times 0.5 \mu\text{m}$.

surfactant-modified PEDOT: PSS film (n-PEDOT: PSS-N) has a conductivity of 0.1 S/cm, lower than that of the pristine PEDOT: PSS film, indicating a negative effect. However, after applying the SVPT, the conductivity of all samples is somewhat improved, where the highest conductivity obtained is 1.7 S/cm, which is 12 times larger than that of the pristine non-vibrated PEDOT: PSS film (0.14 S/cm), implying that the SVPT method combined with surfactant modification has a great potential to fabricate thin films with high conductivity.

In order to further unveil the mechanism of operation of the SVPT method on the film structure, we obtained the AFM phase images of the m-PEDOT: PSS films fabricated with and without applying SVPT (c.f. Fig. 8a). The film microscopic structure, morphology and the distribution of the phases significantly affect the film electrical properties [29]. For the pristine non-vibrated PEDOT: PSS film (PEDOT: PSS-N film), the phase separation between PEDOT and PSS is not evident, which can be due to the commonly accepted opinion that the surface of the PEDOT: PSS film contains more of the PSS phase [16], due to the amphiphilic nature of PSS that causes the accumulation of PSS on the surface [24]. When pristine PEDOT: PSS film is modified with anion and cation surfactants, i.e. the samples a-PEDOT: PSS-N and c-PEDOT: PSS-N, the phase separation between PEDOT and PSS becomes discernible, in which the bright regions can be regarded as the relatively hard phase (PEDOT), while the dark regions can be attributed to the relatively soft phase (PSS and surfactants) [30]. This is also in-line with the conductivity values from Fig. 7. When the non-ionic surfactant is used, no clear phase separation occurs, in accord with a low conductivity in n-PEDOT: PSS-N sample. After applying the SVPT, the phase separation between PEDOT and PSS/surfactant becomes distinguishable, showing more bright regions

in the phase images. This can be explained as the attraction forces between PEDOT and PSS chains are weak Coulomb forces [11]. After imposing the ultrasonic vibration on the film, the Coulomb forces between PEDOT and PSS chains are likely to break off resulting in phase separation. Moreover, the phase images of those samples treated by the SVPT show that the ultrasonic energy of the vibration can contribute to the localized ordering of PEDOT and PSS chains. The arrows and squares in Fig. 8a show the ordering direction of the phases in the selected regions, indicating that the ultrasonic energy forces the PSS chains to reorganize and the PEDOT chains to stick together to form localized conducting paths.

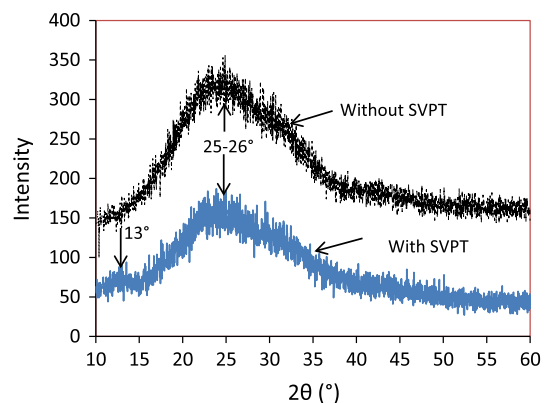


Fig. 9. XRD patterns of pristine spun-on PEDOT: PSS films made with and without substrate vibration. An additional peak has appeared for the sample treated by SVPT, indicating separation of PEDOT and PSS as a result of vibration. Two peaks are observed for the PEDOT: PSS film treated by SVPT, indicating phase separation as a result of substrate vibration.

This ordering effect has been proved to be able to enhance the charge carrier transport and the number of polarons or bipolarons, giving rise to an increase the conductivity [31]. Therefore, based on the above discussion, the mechanism of SVPT is schematically depicted in Fig. 8b. Initially, the PEDOT and PSS chains are randomly arranged in the m-PEDOT: PSS films. After applying the SVPT, the PEDOT and PSS chains become phase separated and ordered in certain areas, improving the overall film conductivity. To further support the argument of phase separation as a result of applying substrate vibration, XRD patterns of pristine PEDOT: PSS fabricated with and without applying vibration are shown in Fig. 9. PEDOT: PSS has a peak around 26° , as observed in both cases. An additional peak has appeared for the sample treated by SVPT at 13° , indicating partial separation of PEDOT from PSS as a result of vibration, in accord with the literature data for peaks of PEDOT that occur around 26° and 13° [32].

4. Conclusion

In this work, we introduced a novel and effective post treatment mechanical method applicable to most film deposition or coating techniques from a solution, such as spin or dip coating. The method is termed as the ultrasonic substrate vibration post treatment (SVPT). In SVPT, a wet film is placed on a planar ultrasonic transducer, immediately after casting, for further treatment. The SVPT is a typical mechanical method which avoids several disadvantages of chemical and thermal post treatment methods, such as high cost and environmental concerns. In this paper, the SVPT was applied on a sample solution, i.e., PEDOT: PSS in the pristine aqueous solution, as well as solutions modified by surfactants. The effect of the substrate vibration duration was studied on the thickness, roughness, and electrical conductivity of pristine PEDOT: PSS films, and it was found that a short duration vibration (10 s, in the range of the parameters studied here including a 5 W ultrasonic vibration power equivalent to a $8.7\ \mu\text{m}$ vertical displacement of the substrate) results in the formation of a thinner, smoother and more conductive film. This is because the ultrasonic energy may be a disruptive or destabilizing effect, implying that only low power and short duration ultrasonic vibration can favor the uniformity of wet films. The conductivity of the best pristine PEDOT: PSS film exhibited four times increase compared with the non-vibrated sample.

We also investigated the effect of vibration on the characteristics of modified-PEDOT: PSS films by adding three different kinds of surfactants to the pristine solution. By comparing the film thickness, roughness, conductivity and phase images of samples made with and without SVPT, it is concluded that substrate vibration post treatment can effectively result in a decrease in the film thickness, roughness and an increase in conductivity. The ultimate results show that the conductivity of spun-on PEDOT: PSS film modified with anion surfactant and fabricated using SVPT method shows an impressive twelve-fold increase compared to that of the pristine non-vibrated PEDOT: PSS film.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ultras.2015.12.012>.

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